POKEMON TCG LIVE: ENTITY RELATIONSHIP DIAGRAM

Software Construction and Decision Making

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1. **Player_account**

Is the entity base in the system and represents the players in the system, the information is used for authentication and player's progression

Attributes: Player_ID, Username, Email, Password, Level, Rank

PK: Player_ID

UNIQUE: Email, Username

NOT NULL: Email, Password

CHECK: Level >= 0, Rank >= 0

2. **Player_wallet**

Disjoin the player's economy with the principal player_account table

Attributes: Wallet_ID, Player_ID, Coins, Crystals, Trainer_points

PK: Wallet_ID

FK (unique): Player_ID

CHECK: Coins >= 0, Crystals >= 0, Trainer_points

3. **Collection**

Store the player's cards that the player owns and make a relationship between player and cards with quantity

Attributes: Player_ID, Card_ID, Quantity

FK (unique): Player_ID, Card_ID

Check: Quantity >= 0

4. **Deck**

A player can create multiple decks in his account. Each deck consists in a group of cards no more than 60 cards

Attributes: Deck_ID, Player_ID, Name, Quantity

PK: Deck_ID

FK: Player_ID

NOT NULL: Name

CHECK: Quantity > 0 && Quantity <= 60

5. **Deck_card**

Relationship between Deck and Card tables, get card design and his attributes and store inside of the deck

Attributes: Deck_ID, Card_ID

FK (unique): Deck_ID, Card_ID

6. **Card**

Centralize all cards exits in the Pokemon TCG Live, stores general attributes shared with all types card exits

Attributes: Card_ID, Name, Card_type, Serie

PK: Card_ID

NOT NULL: Name, Card_type

CHECK: Card_type IN (“pokemon”, “trainer”, “energy”)

7. **Pokemon_card**

Especializes card table for cards type Pokemon, contains attributes focused on Pokemon where are used as main card in the game

Attributes: Card_ID, Pokemon_type, HP, Stage, Weak, Resistance, Retreat

PK/FK: Card_ID

NOT NULL: HP, Pokemon_type, Stage, Retreat

CHECK: HP >= 0

8. **Trainer_card**

Represents trainer cards which have a different logic compared to Pokemon Cards. Exits 3 types of trainer cards which are: Stadium, Supporter and Item. Each one have differents effects in the gameplay

Attributes: Card_ID, Trainer_type, Description

PK/FK: Card_ID

NOT NULL: Trainer_type

9. **Energy_card**

Represents energy cards with own attributes

Attributes: Card_ID, Energy_type, Special_text

PK/FK: Card_ID

NOT NULL: Energy_type

10. **Attack**

A pokemon card can have multiple attacks, each attack is defined here

Attributes: Attack_ID, Card_ID, Name, Damage, Energy_cost, Description

PK: Attack_ID

FK: Card_ID

NOT NULL: Name, Energy_cost

CHECK: Damage >= 0

11. **Ability**

Models passive abilities of the pokemon card, not all Pokemons have abilities

Attributes: Ability_ID, Card_ID, Name, Description

PK: Ability_ID

FK: Card_ID

12. **Match**

Represent a match and attributes, each match hosts 2 players and have a game format which are extended or standard

Attribute: Match_ID, Game_Mode, Format, Start, End, Result

PK: Match_ID

NOT NULL: Format, Game_mode

CHECK: Start < End

13. **Match_player**

Relates player, player deck and match. Consists in a player participation in a match with his deck

Attribute: Player_ID, Deck_ID, Match_ID

FK: Player_ID, Deck_ID, Match_ID

14. **Match_card**

Controls which card is in the game is present during the game. Each card has a zone

Attributes: Player_ID, Match_ID, Card_ID, Zone_ID

FK: Match_ID, Player_ID, Card_ID, Zone

Check: Zone IS NOT NULL

15. **Zone**

Define game zones (Active pokemon, Benched pokemon, Discard pile, Deck, Prize Cards)

Attributes: Zone_ID, Name

PK: Zone_ID

UNIQUE: Name

16. **Turn**

Controls the time flow of the game

Attributes: Turn_ID, Match_ID, Player_ID, Turn_number

PK: Turn_ID

FK: Match_ID, Player_ID

Check: Turn_number > 0

Relationship	Cardinalit	Brief Description
Card — Pokemon_card	1 : 1	Each Pokémon card is a specialization of a base card
Card — Trainer_card	1 : 1	Each Trainer card derives from a general card
Card — Energy_card	1 : 1	Each Energy card is a subtype of Card
Card — Attack	1 : N	A Pokémon card can have multiple attacks
Card — Ability	1 : N	A Pokémon card can have multiple abilitie
Player_account — Player_wallet	1 : 1	Each player has exactly one associated wallet
Player_account — Collection	1 : N	A player can own multiple cards in their collection
Card — Collection	1 : N	A card can belong to multiple players' collections
Player_account — Deck	1 : N	A player can create multiple decks
Deck — Deck_card	1 : N	A deck contains multiple card entries
Card — Deck_card	1 : N	A card can appear in multiple decks
Deck — Card (via Deck_card)	N : M	Many-to-many relationship between decks and cards

Player_account — Match_player	1 : N	A player can participate in multiple matches
Match — Match_player	1 : N	A match includes multiple participating players
Deck — Match_player	1 : N	A deck can be used in multiple matches
Player_account — Match	N : M	Players participate in multiple matches
Match — Turn	1 : N	A match consists of multiple turns
Player_account — Turn	1 : N	A player can have multiple turns across matches
Match — Match_card	1 : N	A match contains multiple cards in play
Player_account — Match_card	1 : N	A player controls multiple cards during a match
Card — Match_card	1 : N	A card can appear multiple times across matches
Zone — Match_card	1 : N	A zone contains multiple cards during gameplay